

Architecture Diagram Explanation (Version 2)

Overview

The Restaurant Customer Analytics Platform is a fully AWS-native analytics solution designed to provide customer behavior insights, Customer Lifetime Value (CLV) calculations, RFM segmentation, churn indicators, loyalty program analysis, location performance metrics, sales trends, and discount effectiveness reporting.

The platform follows a Medallion Architecture pattern consisting of Bronze, Silver, and Gold layers and uses AWS Glue with PySpark for ingestion, validation, transformation, and business metric generation.

The final business insights are delivered through an interactive Streamlit dashboard.

1. Source System Layer

The source system is a Microsoft SQL Server database containing restaurant operational data.

Source Tables:

- order_items
- order_item_options
- date_dim

These tables contain transactional order data, item customizations, loyalty information, and calendar attributes required for business reporting.

The source system serves as the system of record for customer transactions and restaurant operations.

2. Ingestion Layer

The ingestion layer uses AWS Glue JDBC connectivity instead of AWS DMS.

Components:

- AWS Glue JDBC Connection
- AWS Glue PySpark Ingestion Job
- S3 Ingestion Logs

The Glue ingestion job establishes a secure JDBC connection to SQL Server and extracts source data on a scheduled basis.

Data is loaded into the Bronze layer every 15 minutes.

Benefits:

- Fully AWS-native ingestion
- PySpark-based processing
- Reduced architectural complexity
- Secure JDBC connectivity
- Scheduled incremental loading

The ingestion process generates audit and execution logs which are stored in Amazon S3.

3. Bronze Layer (Raw Data)

The Bronze layer stores raw source data exactly as received from SQL Server.

Datasets:

- order_items
- order_item_options
- date_dim

Characteristics:

- Immutable storage
- Full historical retention
- Encrypted using AWS KMS
- Versioned storage
- Recovery and replay capability

The Bronze layer serves as the foundation of the data lake and provides a trusted copy of all incoming source data.

4. Data Validation Layer

AWS Glue Data Quality validates all incoming data before it is promoted into the Silver layer.

Validation checks include:

- Schema Validation
- Null Value Checks
- Duplicate Record Detection
- Referential Integrity Validation

- Data Range Validation

Validation results are stored in a dedicated S3 Audit bucket.

Benefits:

- Early detection of data quality issues
 - Improved trust in business metrics
 - Auditability and governance
 - Reduced downstream processing errors
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5. Silver Layer (Curated Data)

The Silver layer contains cleansed, standardized, and conformed datasets.

Examples:

- customer_orders
- customer_daily_metrics
- loyalty_metrics
- location_metrics

Transformations performed:

- Data standardization
- Business rule application
- Deduplication
- Conformed dimensions
- Data enrichment

Data is stored as Parquet files and partitioned for performance and cost optimization.

The Silver layer acts as the reusable business data foundation for analytical processing.

6. Gold Layer (Business Analytics)

The Gold layer contains business-ready analytical datasets optimized for reporting and dashboard consumption.

Key Datasets:

customer_ltv_daily

Tracks daily Customer Lifetime Value evolution.

customer_rfm

Stores Recency, Frequency, and Monetary scores.

customer_churn

Tracks customer inactivity and churn indicators.

loyalty_metrics

Measures loyalty program effectiveness.

location_performance

Ranks restaurant performance by revenue and operational metrics.

sales_trends

Provides daily, weekly, and monthly trend analysis.

discount_analysis

Measures promotion and discount effectiveness.

Characteristics:

- Aggregated metrics
- Optimized for analytics
- Business-friendly structures
- Parquet storage
- Encrypted using AWS KMS

7. Metadata and Analytics Layer

The metadata and analytics layer enables discovery and querying of Gold datasets.

Components:

AWS Glue Crawlers

Automatically scan and identify datasets.

AWS Glue Data Catalog

Provides centralized metadata management.

Functions:

- Schema management
- Table definitions
- Metadata governance
- Data discovery

Amazon Athena

Provides serverless SQL access to Gold datasets.

Benefits:

- No infrastructure management
 - Fast analytical querying
 - Cost-efficient reporting
 - Direct integration with Streamlit
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8. Dashboard Layer

Business users access insights through a Streamlit dashboard.

Dashboard Modules:

Customer Segmentation

RFM-based customer grouping.

Customer Lifetime Value

Daily CLV trends and customer value analysis.

Churn Risk Dashboard

Identification of inactive and at-risk customers.

Sales Trends & Seasonality

Weekly, monthly, and holiday trend analysis.

Loyalty Program Impact

Comparison of loyalty versus non-loyalty customer behavior.

Location Performance

Restaurant ranking and operational performance analysis.

Pricing & Discount Effectiveness

Impact analysis of discounts and promotions.

The dashboard enables self-service analytics for business stakeholders.

9. Operations and Governance

The platform incorporates enterprise-grade security, monitoring, and governance controls.

AWS KMS

Encryption of data at rest.

AWS Secrets Manager

Secure storage of SQL Server credentials and secrets.

IAM Roles

Least-privilege access control.

Amazon CloudWatch

Centralized logging, monitoring, and metrics collection.

Amazon SNS

Automated alerting and notification services.

AWS Glue Data Catalog

Metadata governance and schema management.

These services ensure operational reliability, security, compliance, and auditability.

10. CI/CD Pipeline

The solution includes a fully automated deployment pipeline.

Components:

- GitHub Repository
- GitHub Actions
- Code Commit
- Lint and Unit Testing
- Build and Packaging
- Glue Job Deployment
- Streamlit Deployment

Pipeline Flow:

Developer Commit → GitHub Actions → Code Validation → Unit Testing → Build Package → Deploy Glue Jobs → Deploy Streamlit → Production Environment

Benefits:

- Automated deployments
 - Consistent releases
 - Reduced operational risk
 - Faster delivery cycles
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11. Key Architectural Benefits

Reliability

- Automated workflows
- Error handling
- Retry mechanisms
- Monitoring and alerting

Performance

- Partitioned datasets
- Parquet storage format
- PySpark optimization
- Athena query acceleration

Security

- KMS encryption
- Secrets Manager
- IAM role-based access
- Secure JDBC connectivity

Governance

- Centralized metadata
- Data quality validation
- Audit logging
- Lineage support

Cost Optimization

- Serverless architecture
- S3 lifecycle management
- Athena pay-per-query model
- Right-sized Glue jobs

Business Value

This architecture enables the organization to calculate Customer Lifetime Value daily, identify high-value and at-risk customers, evaluate loyalty program effectiveness, optimize promotional strategies, improve operational decision-making, and support executive reporting through a scalable, secure, and cost-efficient AWS-native analytics platform.